

Much of this was laid out in a 2002 paper called “The Denominator Blindness Effect: Accident Frequencies and the Misjudgment of Recklessness.”¹²⁵ Lacking the proper context led to disproportionate judgments.

Dan Gardner, in his book *Risk: Why We Fear the Things We Shouldn’t—and Put Ourselves in Greater Danger*,¹²⁶ calls this practice “Fear, Inc.” There are many industries that thrive on unfounded fears and are only too happy to sell you things you don’t need to keep you safe from criminals, terrorists, germs, the end-of-times, hyperinflation, and other outlier calamities.

When we fear things because they seem scary, but in fact they are highly unlikely, that can lead to poor decisions relating to how much risk we should assume. In Darwinian evolution, those who ignore existential risk don’t get to pass their genes along. Perhaps that’s the reason why we are so instinctually influenced by scary events and large numbers. More than simple innumeracy, it is insidious because we are often unaware of its subtle impact on our psychology.

Denominator blindness is the scourge of anyone who has even the most rudimentary understanding of mathematics in the real world. Anytime someone trots out a big number, make sure you understand the context—and always look for the denominator.¹²⁷



Next up, a few examples of how people take advantage of our denominator blindness.

¹²¹ National Consortium for the Study of Terrorism and Responses to Terrorism (START), University of Maryland.

¹²² Garrick Blalock, Vrinda Kadiyali, Daniel H. Simon, “Driving Fatalities After 9/11: A Hidden Cost of Terrorism,” [researchgate.net](https://www.researchgate.net) (December 5, 2005).

¹²³ Insurance Institute for Highway Safety / Highway Loss Data Institute, “Fatality Facts 2022: Yearly Snapshot,” [iihs.org](https://www.iihs.org) (June 2024).

¹²⁴ CDC Centers for Disease Control and Prevention, Heart Disease, “Heart disease in the United States,” [cdc.gov](https://www.cdc.gov). See also: Max Roser, “Causes of death globally: what do people die from?” Our World in Data (December 7, 2021).